TECHNICAL REPORT

Predicting the Working Life of Algorithmically Discovered Trading Strategies

A survival-analysis study, built and verified against synthetic ground truth

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Repository	survival-analysis-pipeline
Run	seed 7, 5,000 strategies, 5 temporal folds
Source of figures	`reports/synthetic/metrics.json`, regenerated by `python scripts/run_synthetic_pipeline.py`

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1. Summary

This report evaluates two survival models fitted to synthetic data drawn at seed 7. It holds 5,000 strategies, each observed from its start date. 93.2% have observed endings and 6.8% are censored, still running when observation stopped. Median observed duration is 91 days. The models predict, from what was on file at the start date, how long each strategy survives.

The like-for-like comparison is the fold mean. Each of the 5 temporal folds fits both models on its own window, and the fold scores average. On concordance, the share of pairs a ranking orders correctly where 0.500 is a coin flip, XGBoost AFT scores 0.781 and the Cox proportional hazards baseline 0.780; the boosted model scores higher.

Pooled over 3,000 out-of-time test strategies, the AFT model scores 0.781 (95% bootstrap interval 0.772 to 0.789); section 5 explains how the two figures differ.

Both fitted models are saved; the bundle records the recommended model for scoring new rows.

Little of the pooled figure is `asset_class` group membership: group means alone score 0.538, within-group comparisons 0.779; section 5 gives the decomposition.

An oracle ranking (Addendum A) bounds every model at 0.820; ranking by validation Sharpe scores 0.410, below the coin flip in every fold (mechanism in the notes).

> All results are synthetic; the run validates the pipeline and asserts nothing about live data.

2. Motivation

An automated strategy search produces a queue of candidates that all clear validation, and they stop working at very different rates. How much capital a new strategy receives, when its first review is scheduled, and when it is retired all depend on how long its edge will last. Treating every deployment the same misallocates in both directions. It overfunds strategies that will not survive the quarter and underfunds ones that would have run for a year.

The question this run poses is whether discovery-time metadata carries enough signal to separate them. A strategy selected from a hundred thousand candidates carries more selection bias than one selected from a hundred, and a strategy whose walk-forward Sharpe was already sliding during validation had begun decaying before it was deployed. None of that is visible in a single validation statistic, and all of it is already recorded. Death here is a bookkeeping event, the date a strategy stops clearing the book's retention rule, so the lifetimes any model learns are partly a property of that rule.

3. Data

Start dates run 2021-07-01 to 2026-06-01; durations are measured in days.

The pipeline has no support for left truncation, strategies whose life began before the source window. Those must be excluded during preparation, and whether they were is recorded with the dataset.

6% of strategies are administratively retired independent of performance, and survival time is log-normal in the latents at scale 0.55.

The generator installs its structure on purpose, so that recovering it is a test rather than an interpretation. Each of the six feature families shifts log survival time by a fixed amount, microstructure the most fragile and value-carry the most durable. Among the four asset classes, crypto is built to decay fastest, a shift of -0.30 on log time against the fx-majors baseline. It is drawn for roughly 15 percent of rows, so the class effect has to be recovered from a minority slice.

Censoring concentrates among recently discovered strategies, which have had the least time to fail before the observation cutoff, so the final fold's test block is far more censored than the population overall.

Figure 1 shows Kaplan-Meier survival curves by `asset_class`, the coarsest structure in the outcome before any model is fitted.

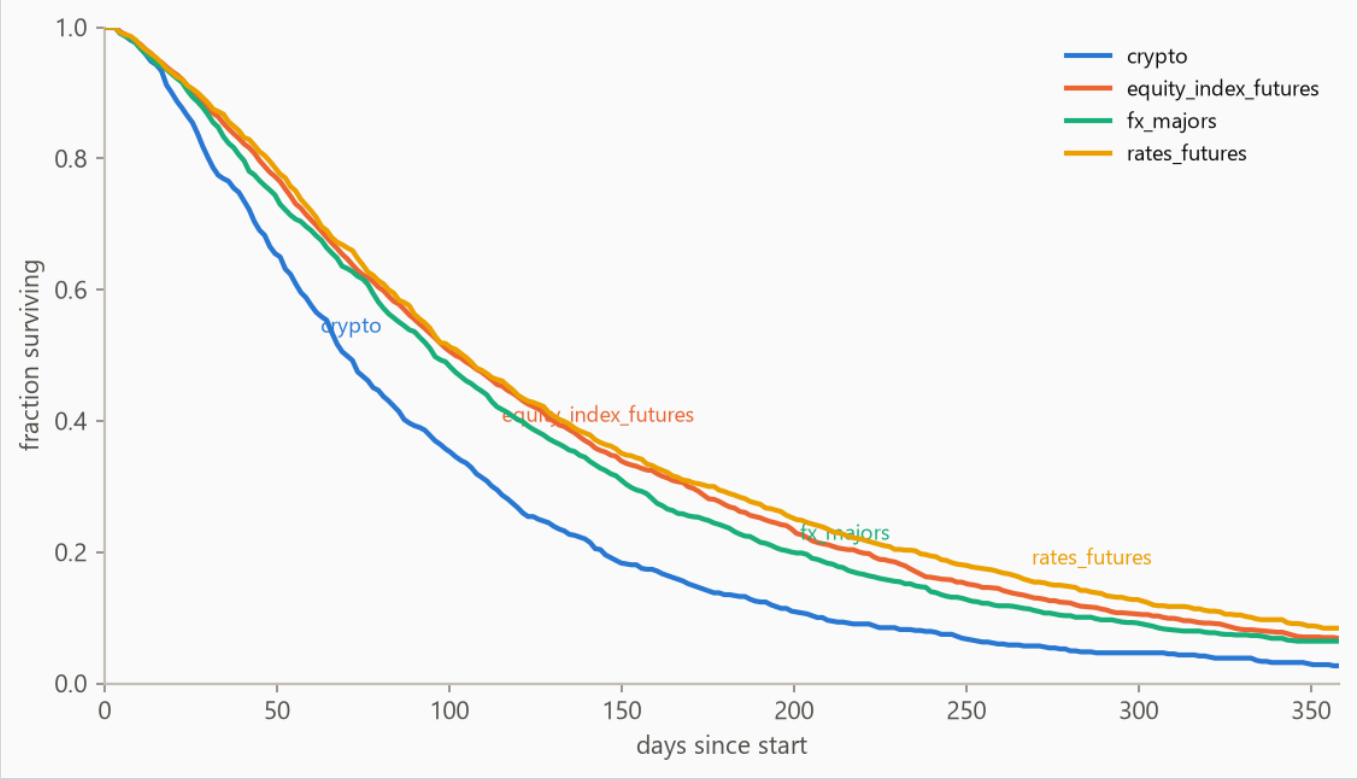

Figure 1. Kaplan-Meier survival curves by `asset_class`: the fraction of each group still running at each age, with censored strategies counted for as long as they were observed. Groups beyond the seven most frequent are collapsed into (other) for this plot only.

4. Method

4.1 Model class

The target is a duration with incomplete observations, so the model is an accelerated failure time (AFT) model, XGBoost with its `survival:aft` objective. An observed ending enters as [t, t] and a censored strategy as [t, infinity), so every strategy contributes. The model predicts each strategy's median survival time in days, and a log-normal curve of fitted, shared width around that median gives the probability of surviving any horizon. A Cox proportional hazards baseline, the standard linear survival model, is fitted with lifelines' `CoxPHFitter` on the same features and answers whether the boosted model was necessary.

Numeric features pass through, missing values included (the Cox baseline gets train-window median imputation); text columns are one-hot encoded with the vocabulary refit per training window, so a fold's features reflect only what was on file by its split date.

4.2 Temporal validation and label re-censoring

Evaluation uses 5 expanding-window folds ordered by start date: the earliest 40% of strategies is burn-in, and each fold trains on every strategy started before its split date and tests on the next block (sizes in Table 2).

Training labels are re-censored at each split date. A strategy started long before a split may have died after it, and its label contains that future, so every post-split death is rewritten as a censoring at the split. Omitting this raises scores by importing the future, which makes it the most consequential detail in the pipeline; it is a standalone function (`cv.recensor`) with dedicated tests.

4.3 Selection and calibration

Hyperparameters are selected once, on the first fold's training window, by an inner temporal split scored on held-out censored log-likelihood; concordance is scale-invariant and cannot see an unusable distribution width. The predictive scale, the reported curves' width, is measured on a temporal tail slice by a probe model that never trained on those rows, then carried over to the model refitted on the full window; the selected loss scale was 0.6, the calibrated predictive scale 0.64.

5. Results

5.1 Discrimination

Table 1. Concordance index by model, pooled over 3,000 test strategies with 95% percentile bootstrap intervals over 500 resamples. Higher is better; 0.500 is a coin flip. Fold-mean rows score each of the 5 folds separately and average. The interval resamples test rows with the fitted models held fixed, so it measures scoring precision, not stability across history; the fold spread is the guide to that.

METHOD	CONCORDANCE (HARRELL'S C)	95% INTERVAL
Oracle ranking on latent log-time (ceiling)	0.820	not resampled
XGBoost AFT (pooled)	**0.781**	**0.772 to 0.789**
XGBoost AFT (fold mean)	0.781	not resampled
Cox proportional hazards (fold mean)	0.780	not resampled
Rank by validation Sharpe	0.410	0.397 to 0.422

Cox risk scores are relative to each fold's own window, which is why fold-mean rows are the like-for-like comparison. The pooled row concatenates 5 separately fitted AFT models whose levels can differ, blurring cross-fold pairs; on this run it matches the fold mean at the printed precision.

Folds 2 and 3 are the weakest, at 0.768 and 0.767, too close to separate.

The pooled concordance decomposes by `asset_class`: group means alone score 0.538, and comparisons restricted to same-group strategies score 0.779 (pair-weighted across 4 qualifying groups). The within-group distance above 0.500 is strategy-level skill; the rest is group membership.

Validation-Sharpe ranking scores 0.410 pooled, 0.354 to 0.427 across folds, never above 0.500. Reversed, it scores 0.590, its arithmetic complement, short of the model's 0.781.

Figure 2 plots the per-fold concordance from Table 2.

Table 2. Per-fold results. Censoring is the share of each fold's test strategies whose ending was not observed.

FOLD	SPLIT DATE	TRAIN N	TEST N	CENSORED	AFT	COX	SHARPE	ORACLE
1	2023-06-22	1,999	600	2%	0.781	0.783	0.425	0.824
2	2024-02-03	2,600	600	2%	0.768	0.769	0.421	0.808
3	2024-09-01	3,199	600	2%	0.767	0.764	0.413	0.805
4	2025-04-03	3,800	600	9%	0.773	0.769	0.427	0.811
5	2025-11-03	4,399	600	39%	0.817	0.815	0.354	0.846

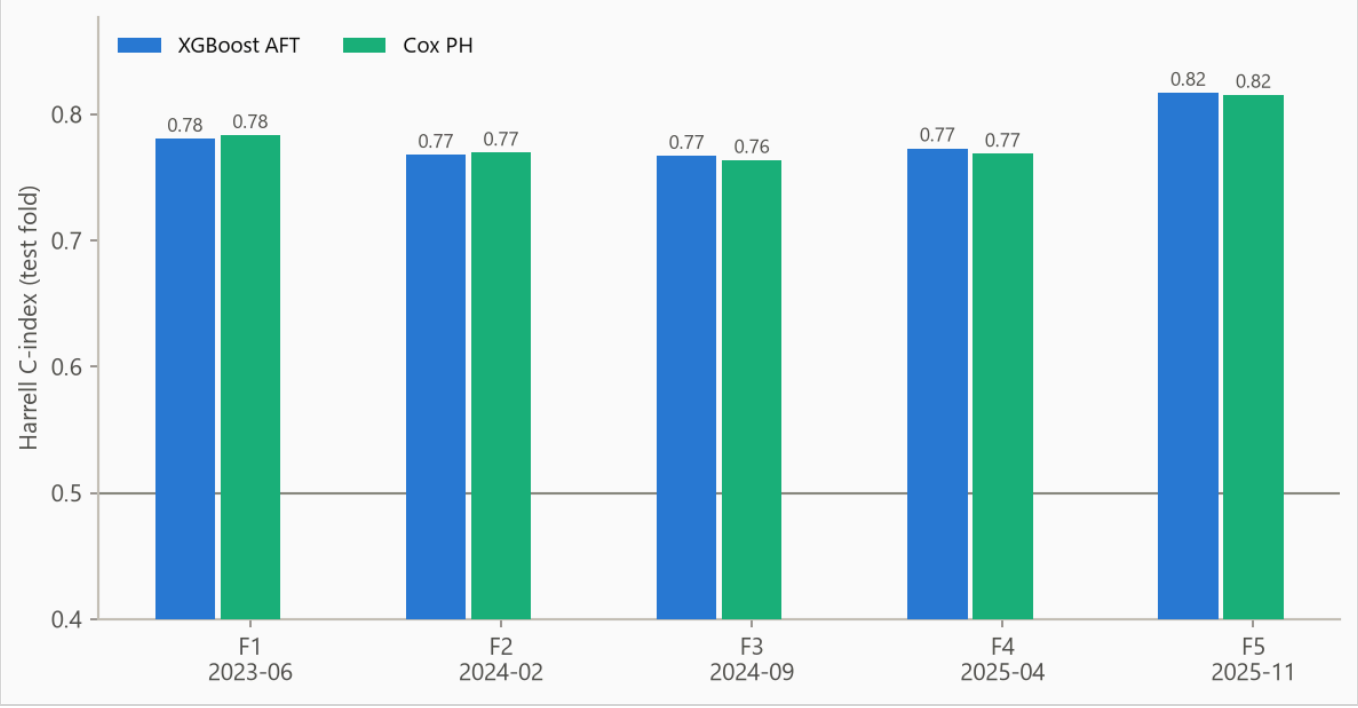

Figure 2. Concordance by fold, from 0.767 to 0.817. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

5.2 Calibration

The Brier score, the mean squared error of a probability forecast (lower is better), is weighted by the inverse probability of censoring (IPCW) so censored strategies do not bias it; the no-skill reference assigns every strategy the population's Kaplan-Meier marginal, the fraction surviving at each age with censored strategies counted while observed.

Table 3. IPCW Brier score by horizon. Lower is better.

HORIZON	AFT	COX	NO-SKILL MARGINAL
90 days	0.149	0.151	0.249
180 days	0.132	0.133	0.182
365 days	0.050	0.051	0.056

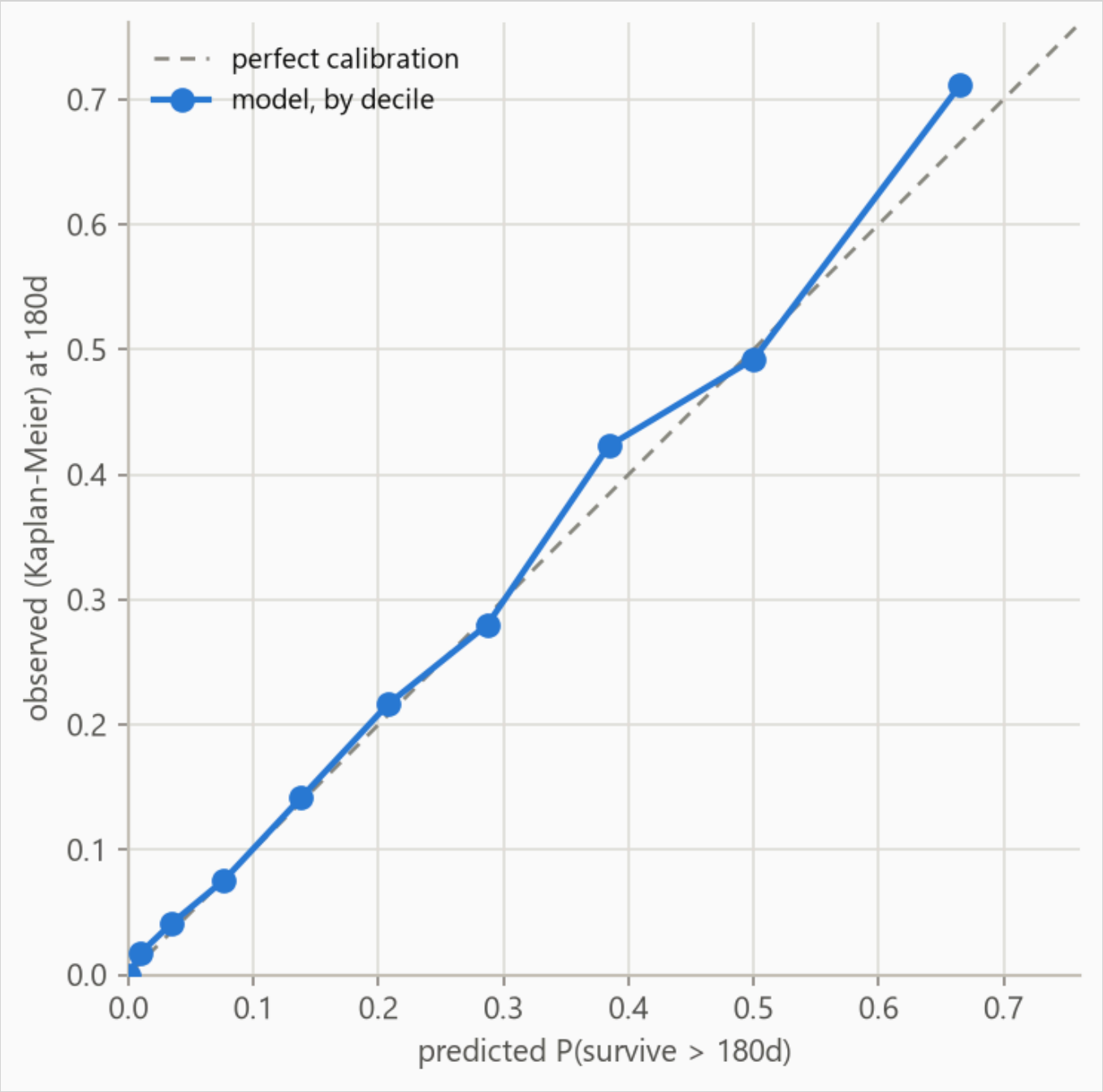

Figure 3. Predicted against observed survival at 180 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored strategies contribute correctly. The largest deviation is 0.046 in decile 10.

Table 4. Decile calibration at 180 days, the values plotted in Figure 3. Deciles are cut on predicted value, so tied predictions make the counts uneven. The observed value in each is a Kaplan-Meier estimate; when a decile's last observed strategy falls short of the 180-day horizon the estimate carries its last value forward, so in small or heavily censored deciles the observed column is softer than its three decimals look.

DECILE	N	PREDICTED	OBSERVED (KM)	DEVIATION
1	300	0.001	0.000	-0.001
2	300	0.010	0.017	+0.008
3	300	0.035	0.041	+0.006
4	300	0.076	0.076	-0.001
5	300	0.138	0.142	+0.003
6	300	0.208	0.216	+0.009
7	300	0.287	0.280	-0.007
8	300	0.384	0.423	+0.039
9	300	0.500	0.493	-0.008
10	300	0.665	0.712	+0.046

6. What the model uses

Feature attributions (SHAP values, for SHapley Additive exPlanations) are computed on the log scale of survival time; +0.3 multiplies predicted survival by about 1.35, and negative values shorten it.

Attributions are descriptive and in-sample, computed on the final model's own training rows. Correlated features split credit by the model's internal choices as much as by the data, so directions are more trustworthy than magnitudes.

Figure 4 ranks features, Table 5 lists the top eight, Figure 5 shows the per-strategy spread, and Figure 6 traces attribution against value.

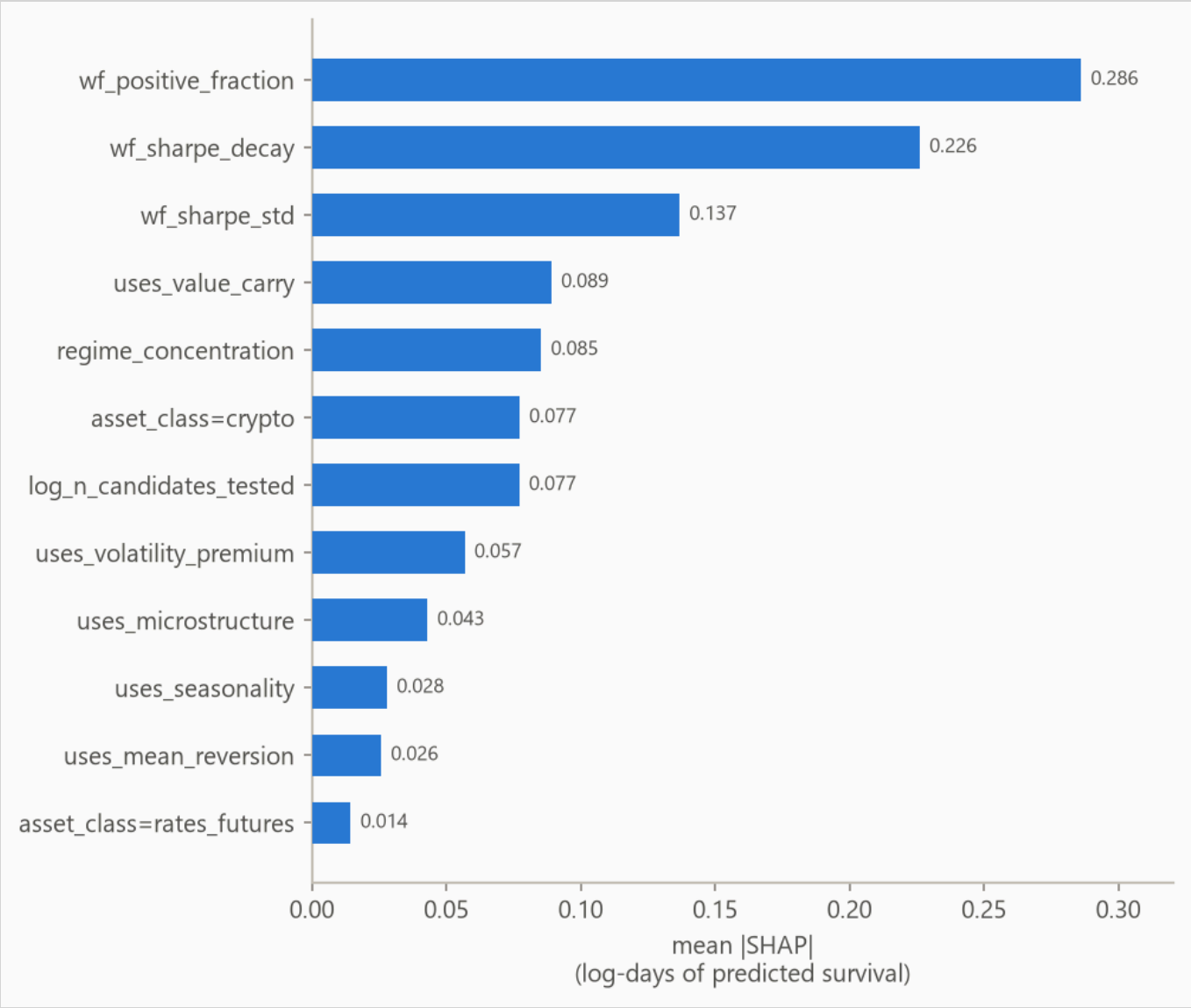

Figure 4. Mean absolute attribution by feature.

Table 5. Top eight features by mean absolute attribution.

FEATURE	MEAN |ATTRIBUTION|
wf_positive_fraction	0.286
wf_sharpe_decay	0.226
wf_sharpe_std	0.137
uses_value_carry	0.089
regime_concentration	0.085
asset_class=crypto	0.077
log_n_candidates_tested	0.077
uses_volatility_premium	0.057

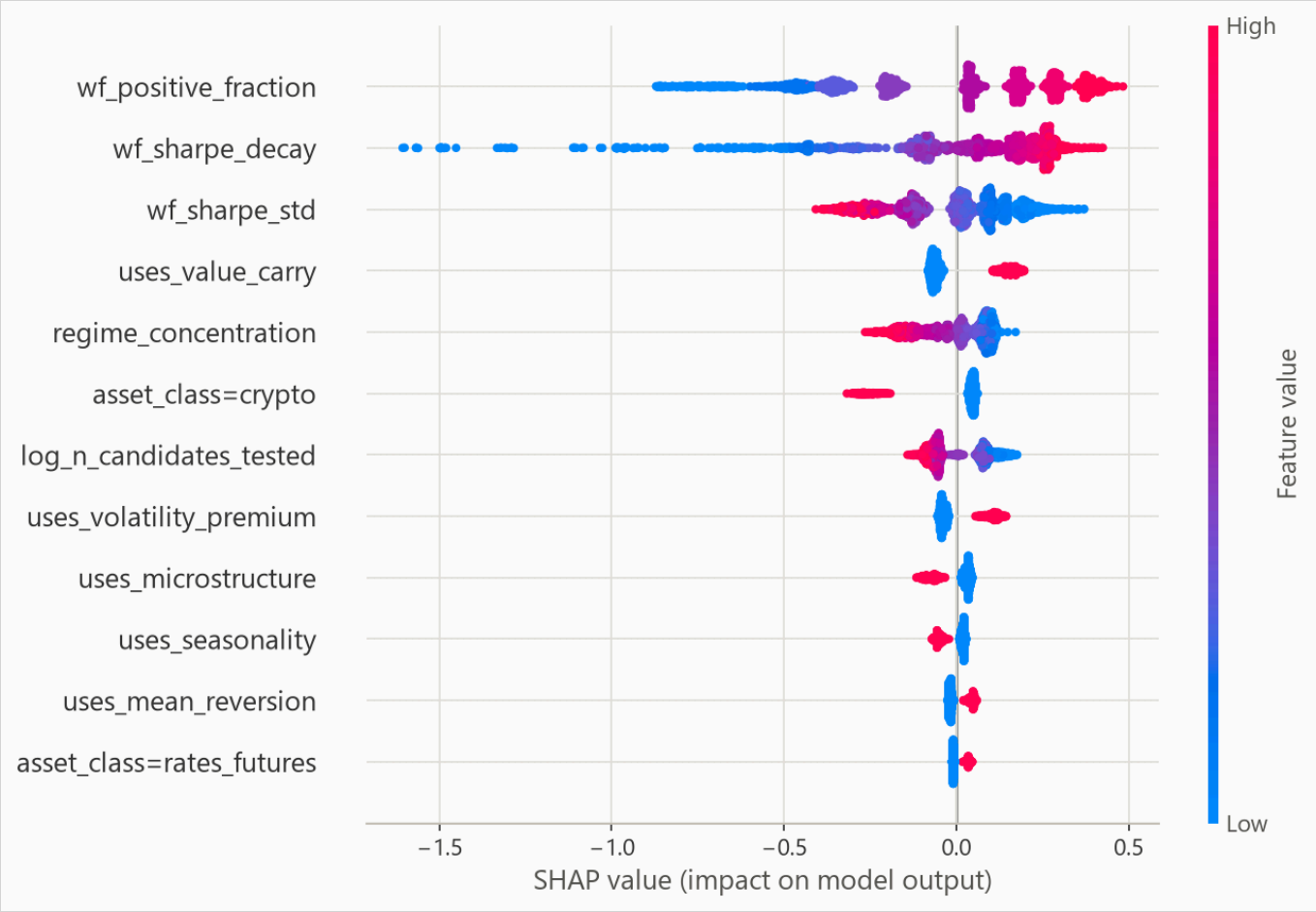

Figure 5. Per-row attributions across the explanation sample. For a yes/no feature, such as a one-hot category flag, the high (red) end simply means the row is in that category.

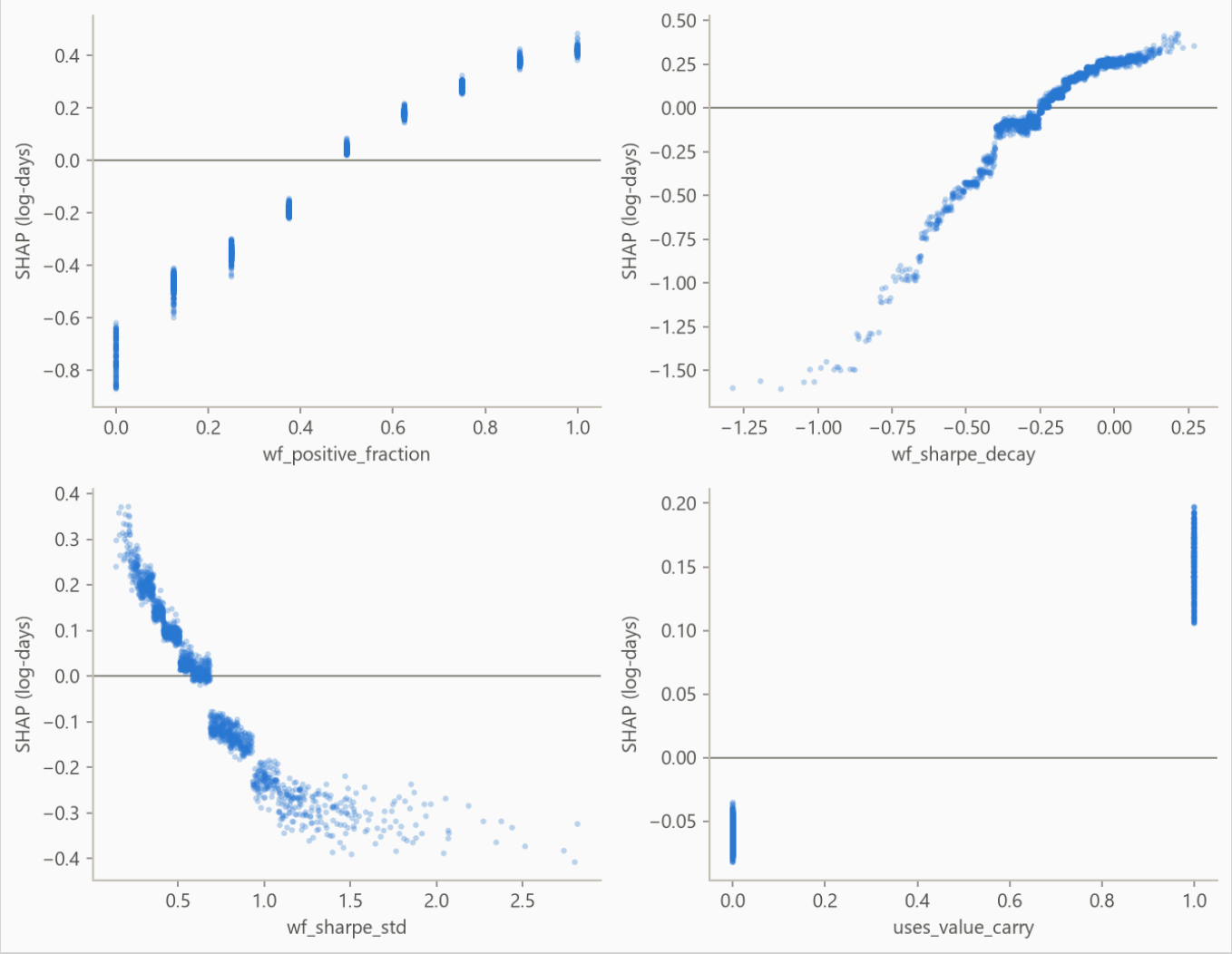

Figure 6. Attribution against feature value for the strongest numeric features; category flags carry no shape and stay in the ranking above. A run short on numeric features fills the grid with flags instead.

7. Interpretation

Why ranking by validation Sharpe inverts. Every strategy entered the population by clearing a validation-Sharpe threshold of 0.8. An observed Sharpe is the sum of true edge, an overfitting component, and noise. Conditioning on that sum exceeding a bar means the survivors with the highest scores are disproportionately the ones whose overfitting and noise broke upward. Past the bar, additional observed Sharpe is more likely inflation than edge. The inflated strategies are the overfit ones, which decay fastest, so higher validation Sharpe actively predicts shorter working life. It scores 0.410 pooled and sits below a coin flip in every fold. This is also what makes the modeling problem worth posing. If the backtest metric ranked survival correctly there would be nothing to add; a meta-model earns its place precisely because selection has already consumed the obvious signal.

Reading the tie. The boosted model and the Cox baseline land within a whisker of each other on the fold mean. The synthetic data generator's observable structure is close to additive, which leaves little for trees to find beyond what a penalized linear model in the log-hazard captures. I report the tie rather than adding interactions to the generator until the headline model wins, because a benchmark tuned until it loses is not a benchmark.

The calibration failure worth keeping. An earlier version of this pipeline selected hyperparameters by concordance and looked correct, then lost to a no-skill forecast on 365-day Brier score. Concordance is invariant to the predictive scale, so the search had settled on a distribution width far too wide and nothing in the training loop objected. That failure is why selection now uses censored log-likelihood and why every report this pipeline produces grades probability quality separately from ranking.

What the attribution shows, checked against the mechanism. The walk-forward statistics that dominate the ranking are not inputs to survival time in the generator. They exist only as noisy proxies for the latent edge-versus-overfit split that is the actual driver, and the model found them and leaned on them. Feature-family and asset-class effects return with the installed signs, among them the crypto shift of -0.30 described in the data note. High walk-forward positive fraction pushes predicted survival up, and a steeply negative walk-forward Sharpe slope pushes it down hard. High fold-to-fold Sharpe dispersion is penalized. All three directions match the generative design, where consistency rises with true edge and falls with overfitting. Validation Sharpe itself is nearly absent from the attribution ranking, which is the winner's curse seen from inside the model. The supportable reading of the magnitudes is that the model uses positive fraction about twice as much as Sharpe dispersion, not that positive fraction is twice as important, since the three walk-forward statistics are correlated proxies of one latent quantity.

Intended use, refit on a real book. The model is a capital-allocation and review-scheduling prior, never a trade signal. Predicted median lifetime sets the first review date and is one input to initial sizing, and the full curve gives a decay schedule to compare live performance against. It prices day-one information only; once a strategy is live, realized performance carries information no discovery-time metadata can. Before any such use, the same temporal cross-validation with the same re-censoring has to be repeated on production metadata, and the resulting concordance should be expected to come in lower, with wider intervals.

8. Limitations

The generator encodes a prior, not evidence, so the concordance here is an upper bound on production performance. Lifetimes are drawn and resampled independently, understating real-book uncertainty, and administrative retirement is modeled as independent censoring, with competing risks left as future work. Every number here comes from one generator seed, so data variance and model variance cannot be separated.

Censoring may be informative. The evaluation assumes strategies stop being observed for reasons unrelated to their risk; early exits of strategies about to end anyway bias survival estimates upward.

Every strategy shares one curve shape. The predictive scale is a single fitted number, so the model shifts a strategy's survival curve earlier or later but never reshapes it; per-strategy width is future work.

Harrell's concordance is biased under heavy censoring, and the final fold is 39% censored; read Table 2 with that in mind.

9. Reproducing this run

Python 3.11 or later. From the repository root:

```
pip install -r requirements.txt
python -m pytest
python scripts/run_synthetic_pipeline.py
```

Every number is read from the run's `metrics.json` at build time, and a figure the prose never cites fails the build. `requirements.txt` pins the exact versions the committed numbers came from.

Addendum A. Synthetic ground truth

Effects are installed as fixed shifts on log survival time; nothing in the generator is proprietary, and the constants are in the run's notes.

The oracle ranking orders strategies by the latent log-time the generator actually used. Nothing is fitted and the ranking predates the noise draw, so its 0.820 bounds every model; the gap to a perfect score is installed noise. A suite test asserts the model never outscores it, since beating perfect information indicates a leak.

Generated from `reports/synthetic/metrics.json` by `scripts/run_build_report.py`. Seed 7, 5,000 strategies, 3,000 out-of-fold test strategies.